



Generative

AI

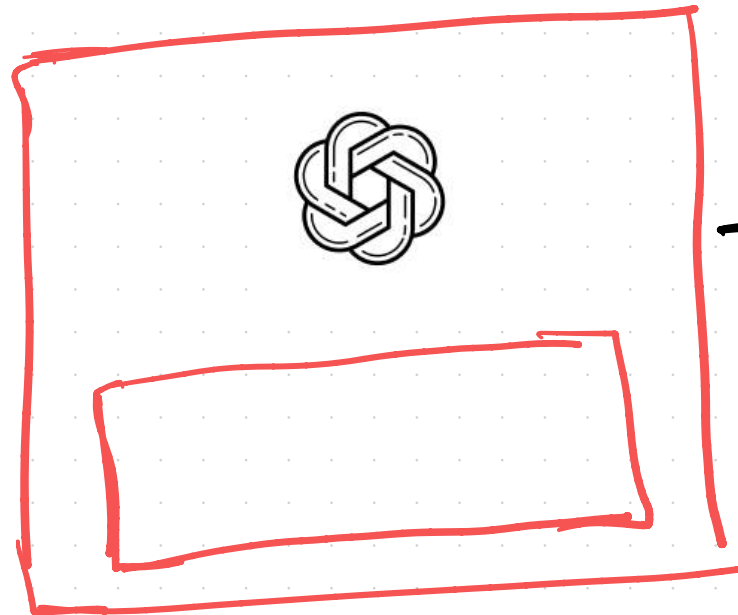


Generative AI
became popular when



launched ChatGPT

Instruction
as a
prompt →



→ text
response

Other types of AI

→ Fraud detection AI

→ Recommender system

→ Image Recognition systems

→ Spam Detection

ML
Models

Traditional AI

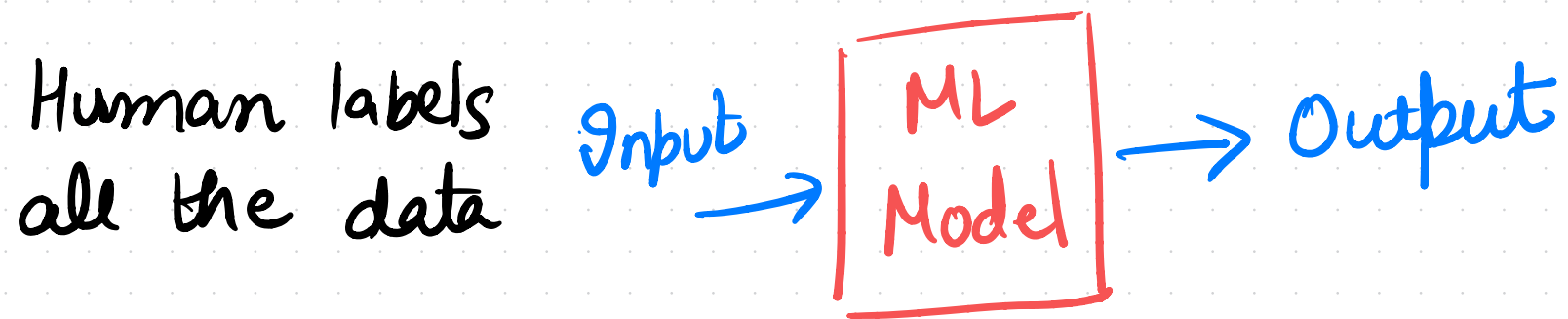


Recognise
cat



Supervised Learning

(learning with a teacher)



- Expensive, slow & limited
- AI can learn only as much humans can label

Self Supervised Learning (without a teacher)



reads book
himself

- ✓ No labels needed
- ✓ AI can learn as much as humans ever recorded

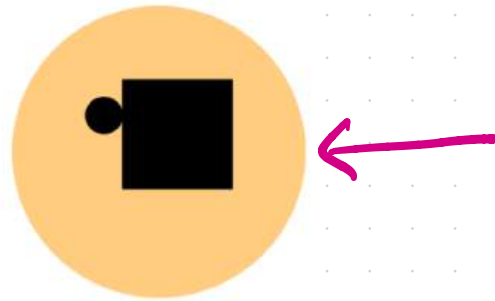
Gen AI



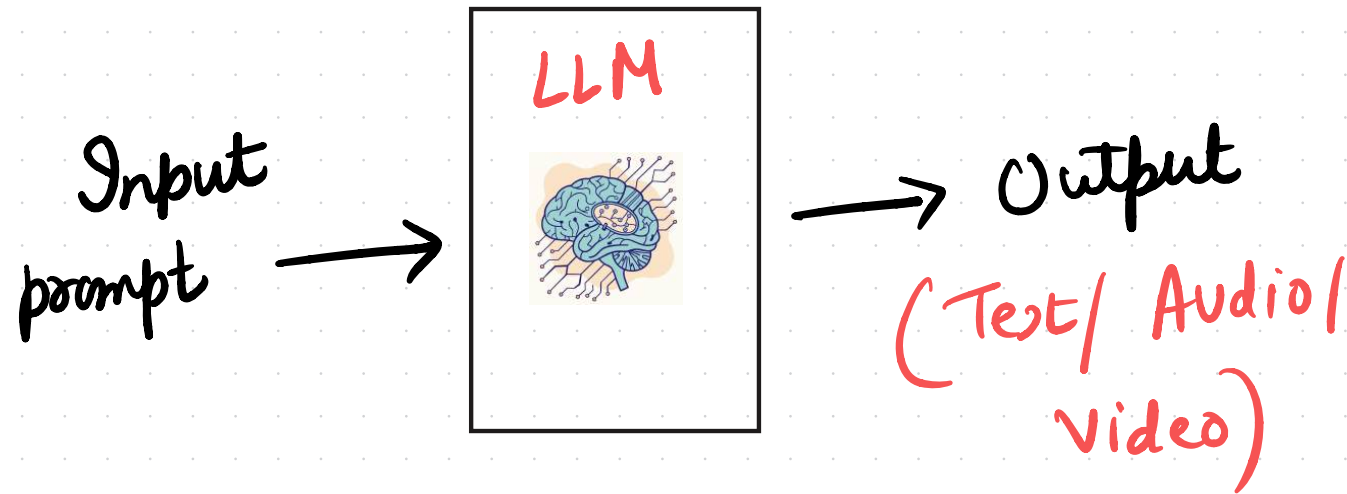
Predicts the missing piece

✓ The cat sat on the ---

✓ Paris is the capital of ---



predict
missing part of
the picture

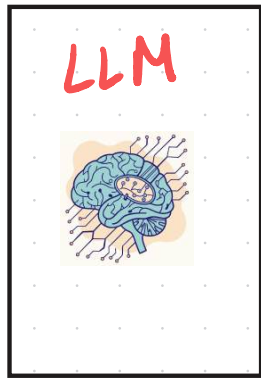


Large → huge training data
huge parameters (weights)

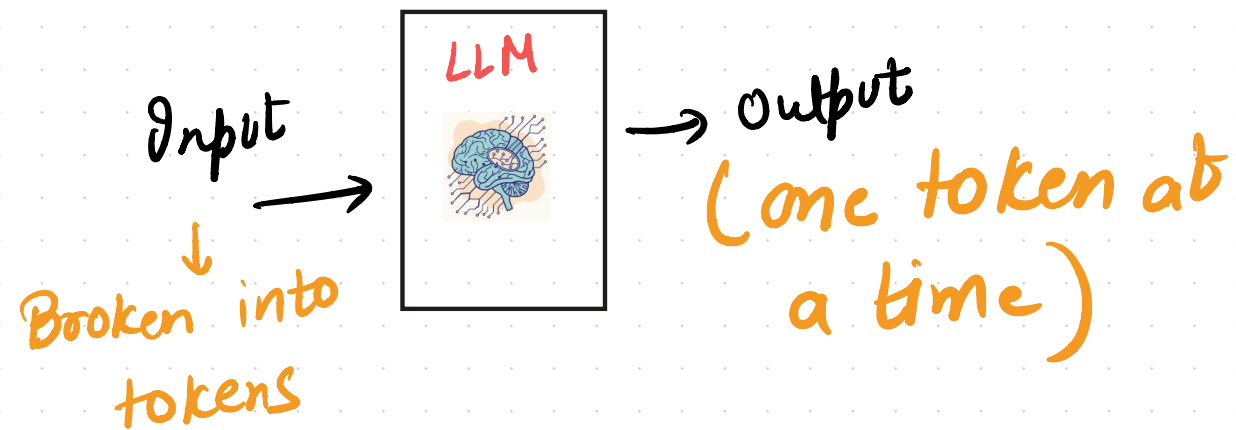
Language → ୧୧ language

Model → Mathematical system to make predictions

Given the previous words, what's the next most likely word to occur

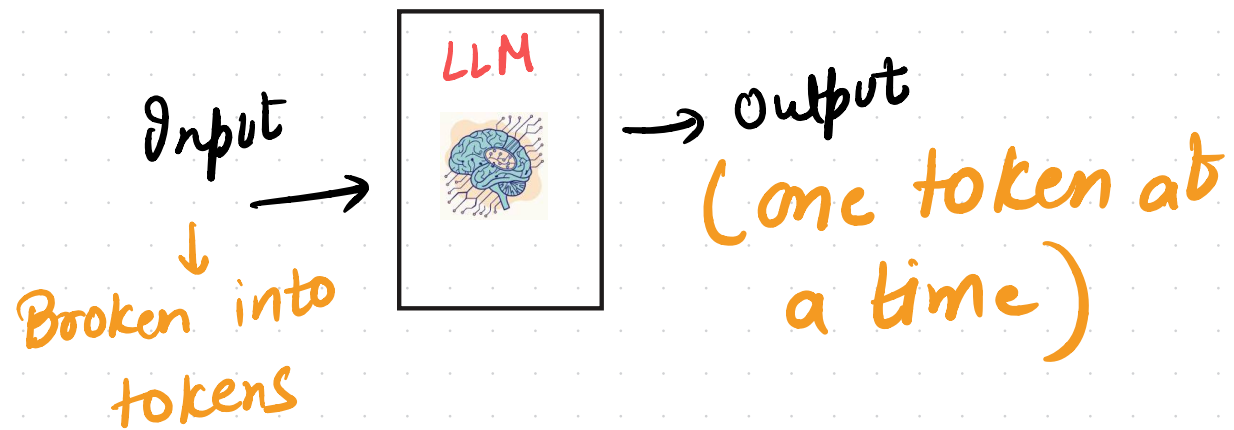


Capital of France is _____ ↙ Paris



Tokens → unit of a text

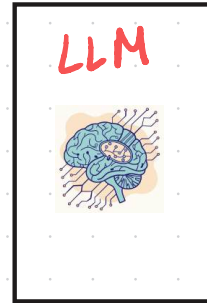
: Once upon a time
unbelievable



↳ Input size is limited based on model context window

Context → working memory of model

Everything that goes inside model



- Your prompt
- entire conversation
-  uploaded
- Instructions
- Model's own prev answers

40

Temporary Chat



Temporary Chat

This chat won't appear in history, use or create memories, or be used to train our models. For safety purposes, we may keep a copy for up to 30 days.

What are some ways to deal with depression? I



Models have no memory between conversations. New conversation cleans up the memory

Every message in the conversation add previous message tokens

Why manage Context?



Model forgets instructions

Output quality degrades



More money

→ You consume more tokens



Slow responses



Pricing

$\textcircled{\$} = \text{Input tokens} + \text{Output tokens}$

↓

Context

[prompt + instruction
+ ---]

↓

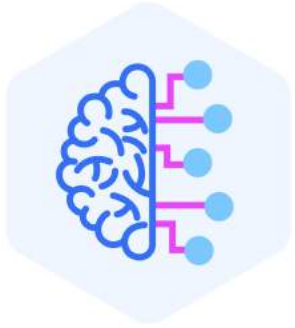
Model Response

$$\text{\$ } 30 + \text{\$ } 70 = \text{\$ } 100$$

Reducing Tokens

Input $\approx$ tokens $\approx$ \$ cost

1. Concise Prompts
2. Limit output length
3. Compress conv history / start new session



Small models
for trivial
tasks



Skills for
repeated
tasks



Scripts for
deterministic
tasks



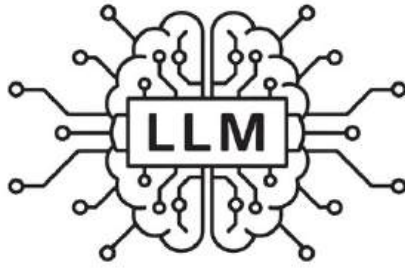
- Opus (Reasoning)
- Sonnet (Balanced)
- Haiku (fast)



- GPT 5.5
- 4.0, 4.1
- 4.0 mini

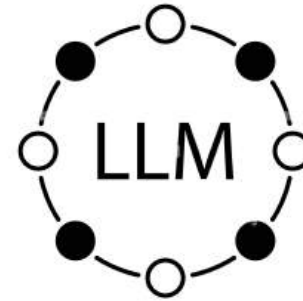


- Ultra
- Pro
- Flash
- Nano



- ✓ Better output
- ✓ Large context

- Slow



- Good for trivial tasks

- Fast

- Cheap

- Hallucinate

? Use Small Models



Draft emails



Fetch JIRA Ticket

? Large Models



Architectural Design



Complex Coding



Plan with large model
Implement with balanced models



Opus for
planning



Sonnet

</>



Haiku
GPT 4o mini



Opus



↓ Create tickets



Create MRs

Trade offs

→ Non deterministic

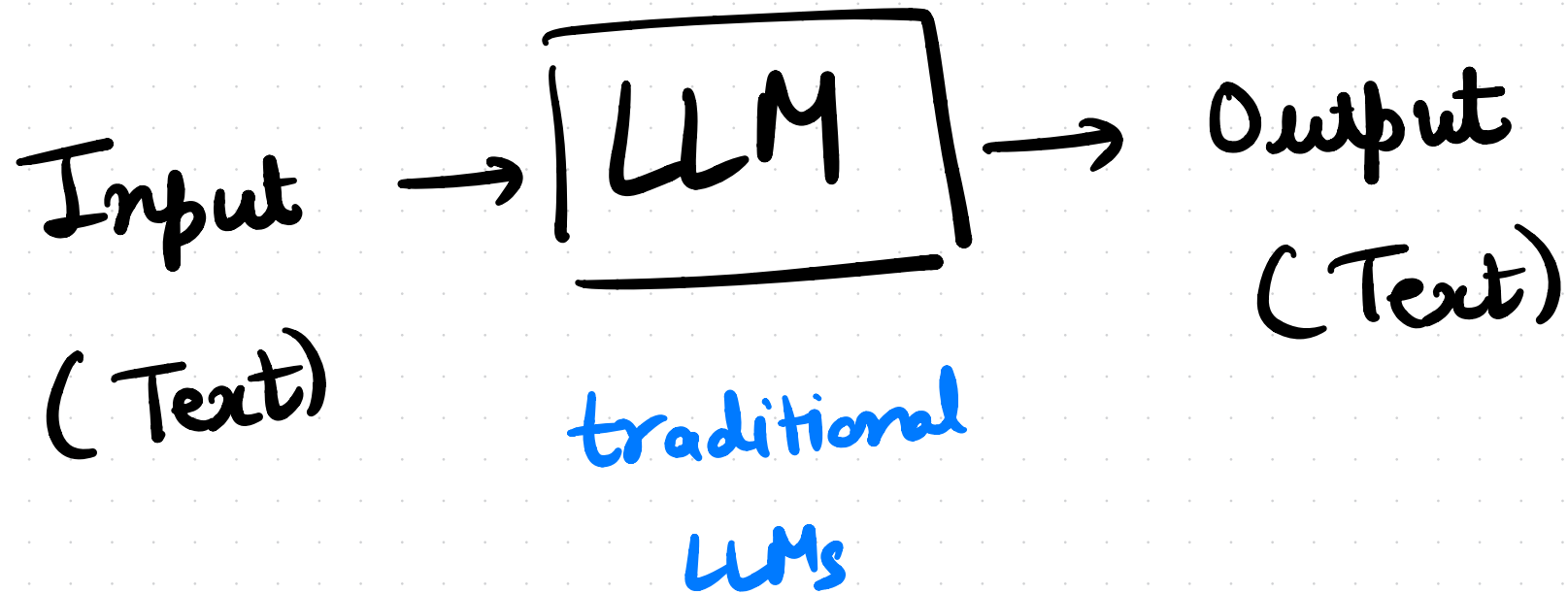
- Best salons in Amsterdam

→ Can hallucinate

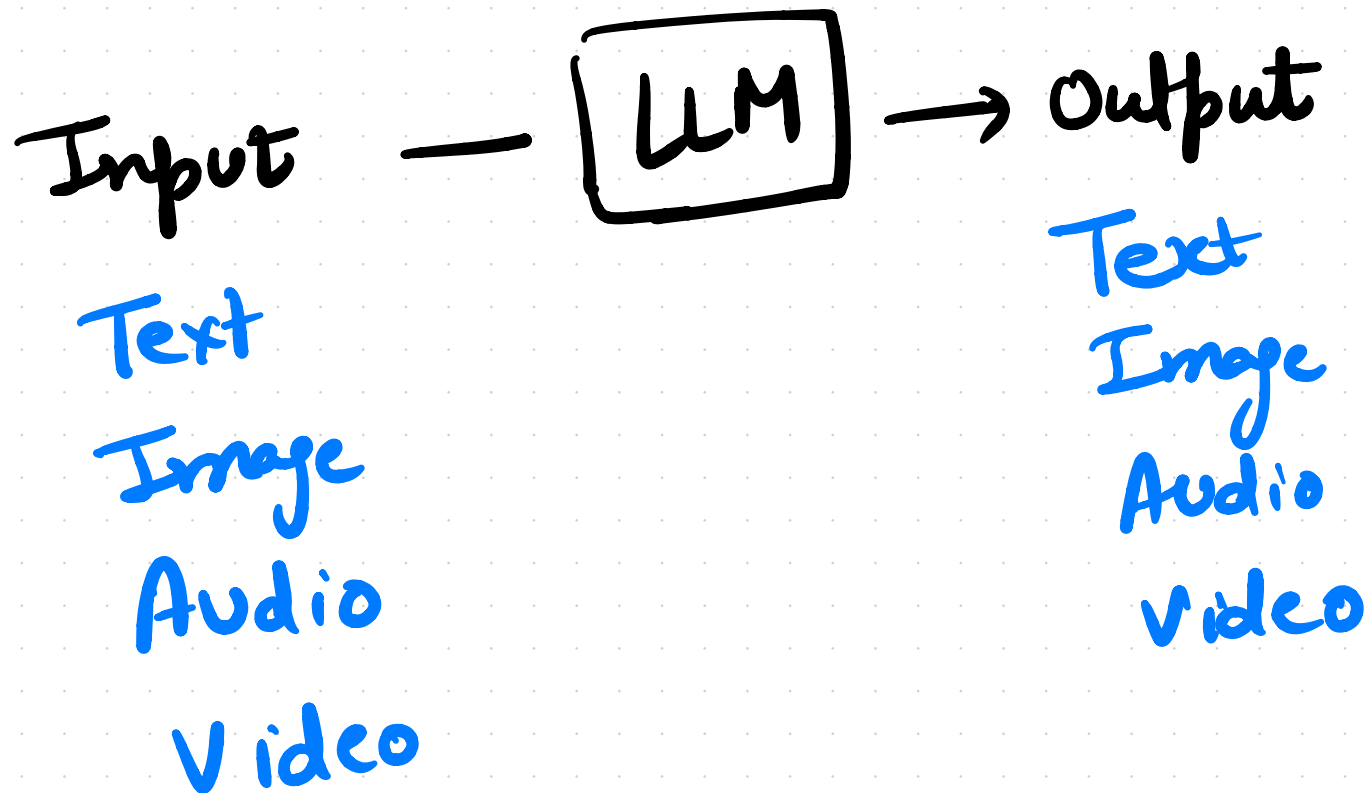
Multimodals

Modalities

- Text
- Image
- Audio
- Video



Advanced LLMs

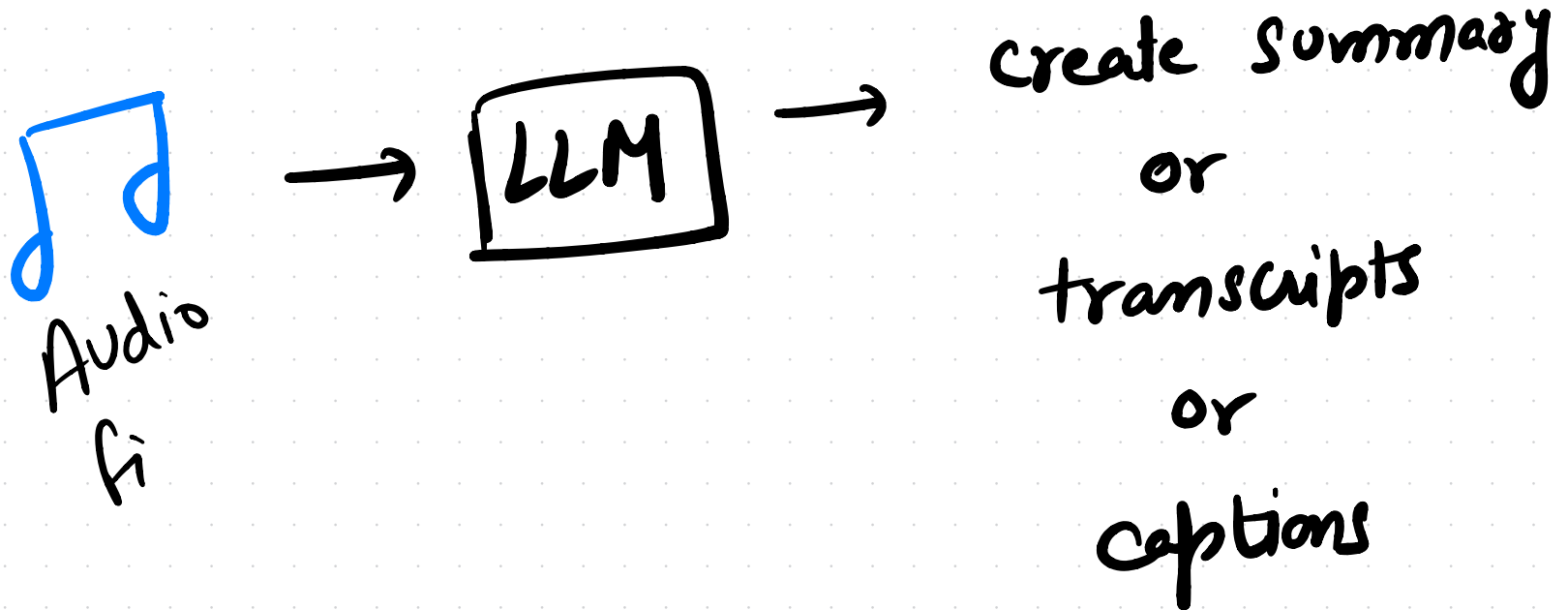


Why?

Because most of the data in
the world is not just

Text

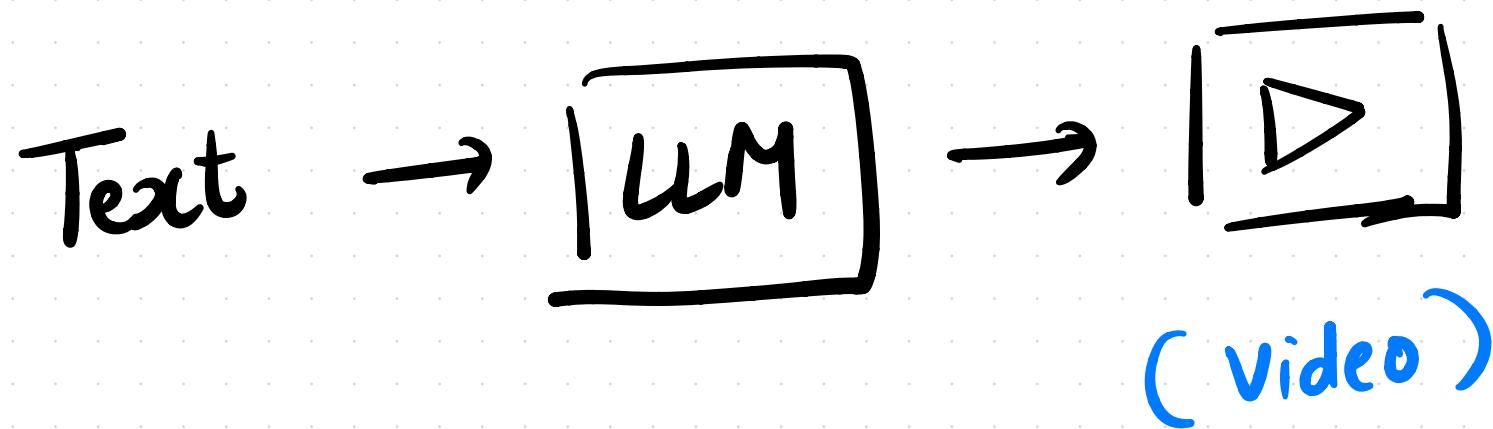
Use Cases



Resolve an Error



Generate a video



We can do alot more
if LLMs can understand
all kinds of data

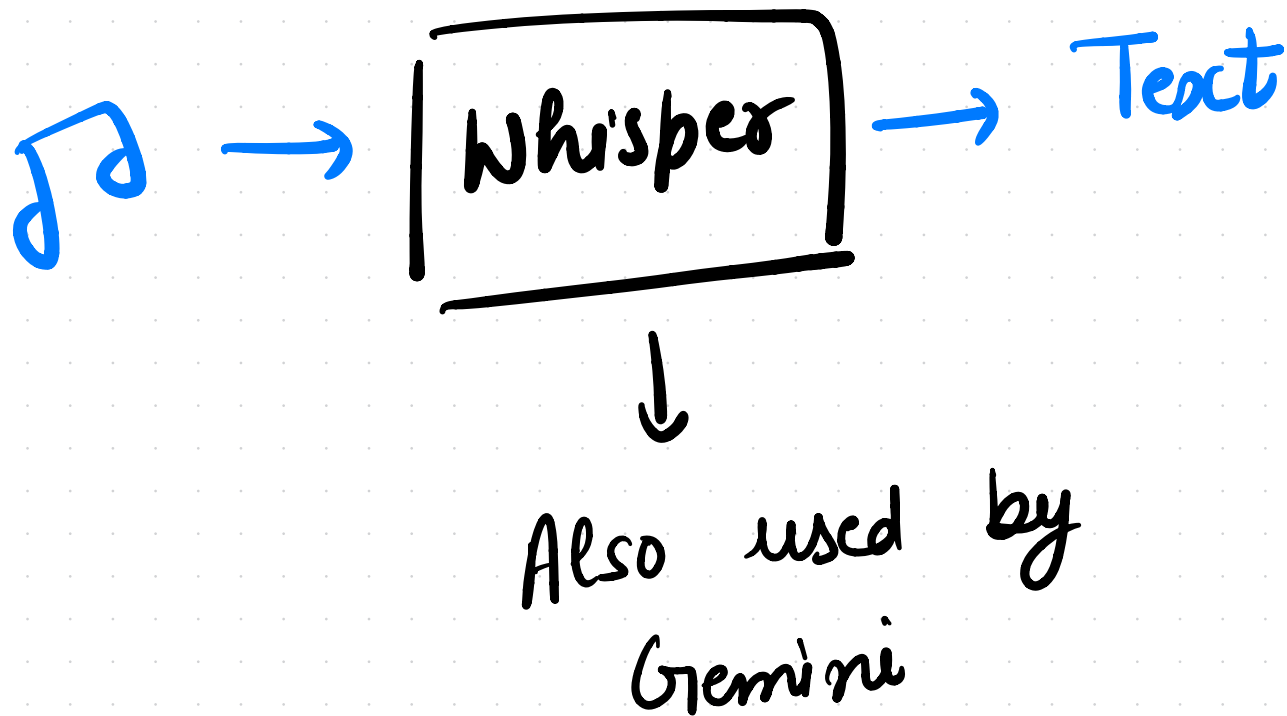
Examples

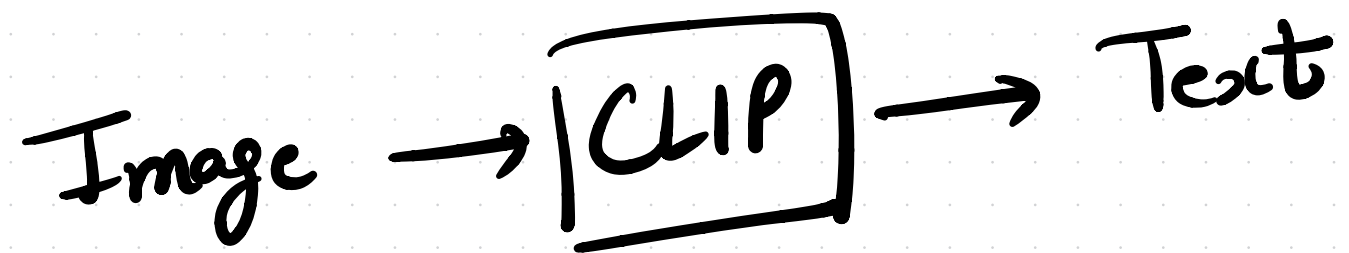
Chat GPT

Claude

Gemini

I/O
(All modalities)





(analysing images)

Vector DB



(Sunset)



Relational DB

Binary
data

Metadata
(Size, name)

tags
(Sunset
beach)

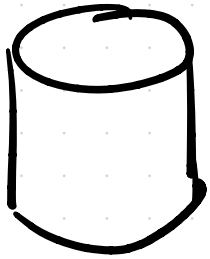


Lacks Semantic context

Warm colors / pink skies

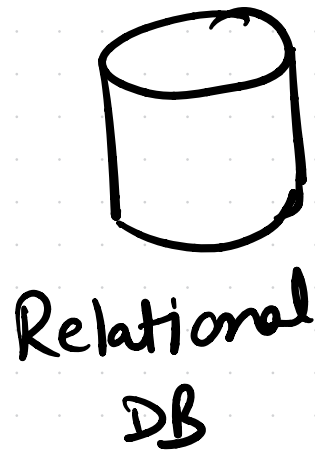
query → 

Get images of same color
palette as the sunset
picture



Relational
DB

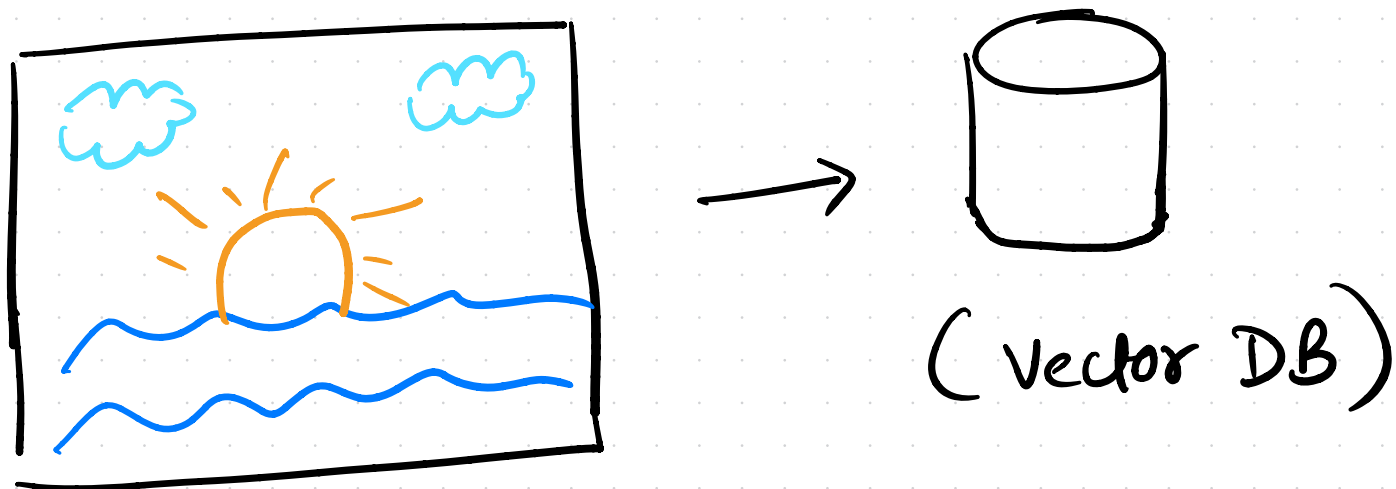
✗
→ not good for
storing unstructured
data



SEMANTIC GAP



Disconnect between how
Computers store data
Vs how humans
understand



Vector dB is used to store
unstructured data like (Images,
Audio, Video)

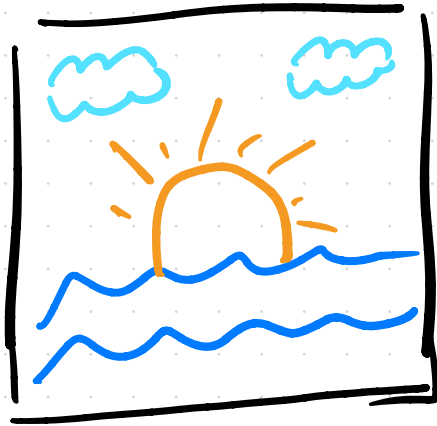
Vector Embeddings

Numerical representation of
the data (Image/Audio/Video)

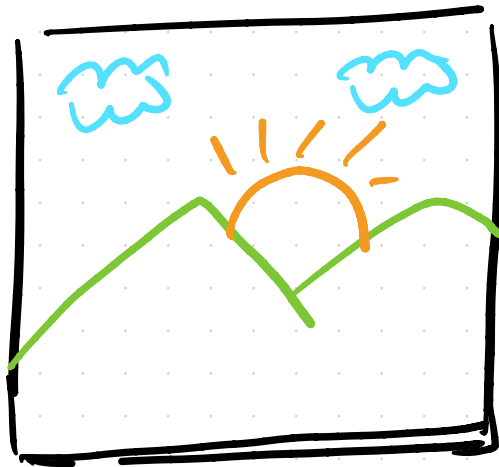
$[0.21, 0.15, 0.83 \text{ ---}]$

↓ ↓ ↓

elevation urban worm colors

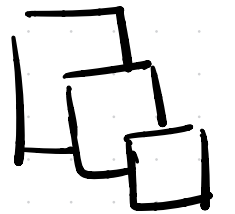


[0.21, 0.15, 0.83 - - -]
↓ ↓ ↓
elevation urban warm
 colors



[0.95, 0.20, 0.42 - - -]

Embedding Models



Data



Vectors

(list of numbers)

Text



Word/sentence embeddings

Image

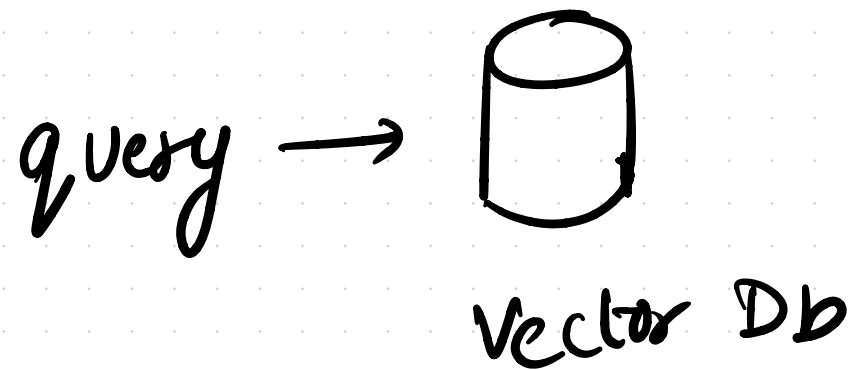
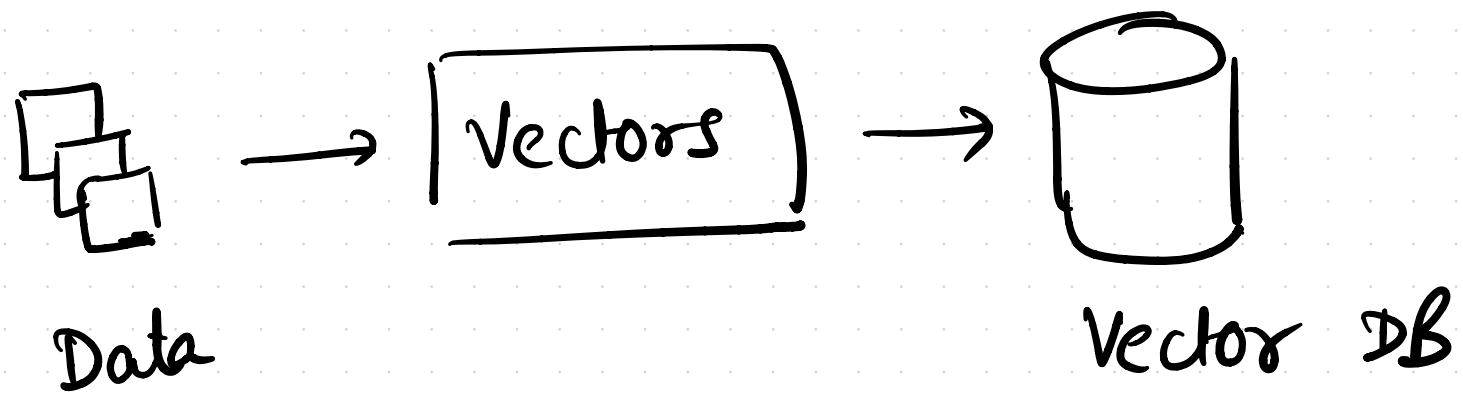


Image embeddings

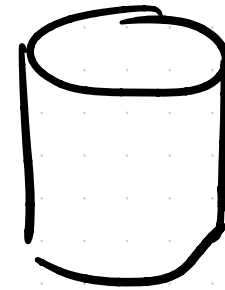
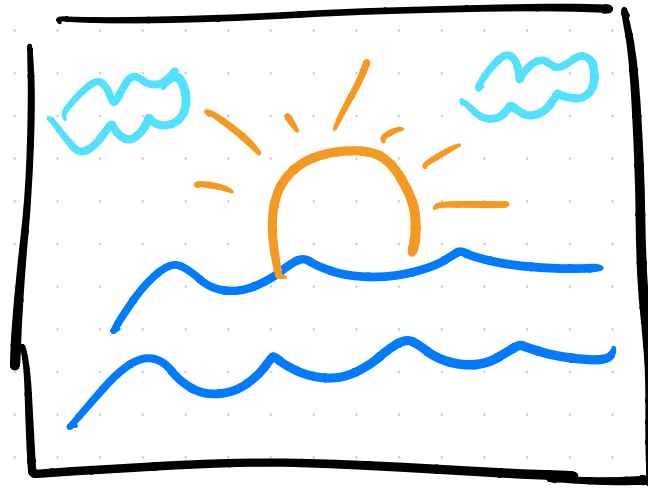
Audio



speech embeddings



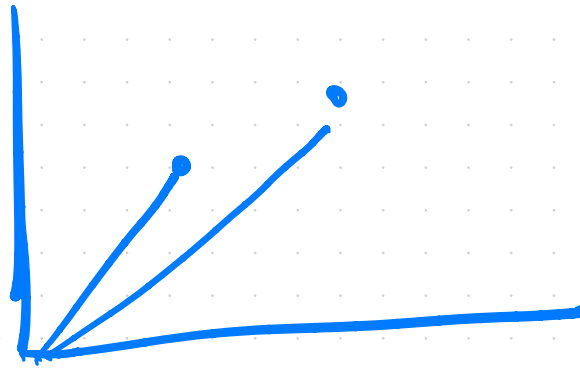
Find exact Match



Vector Db

~~too slow~~

Vector Indexing



ANN algorithm is used to
find the closest match

Use Cases

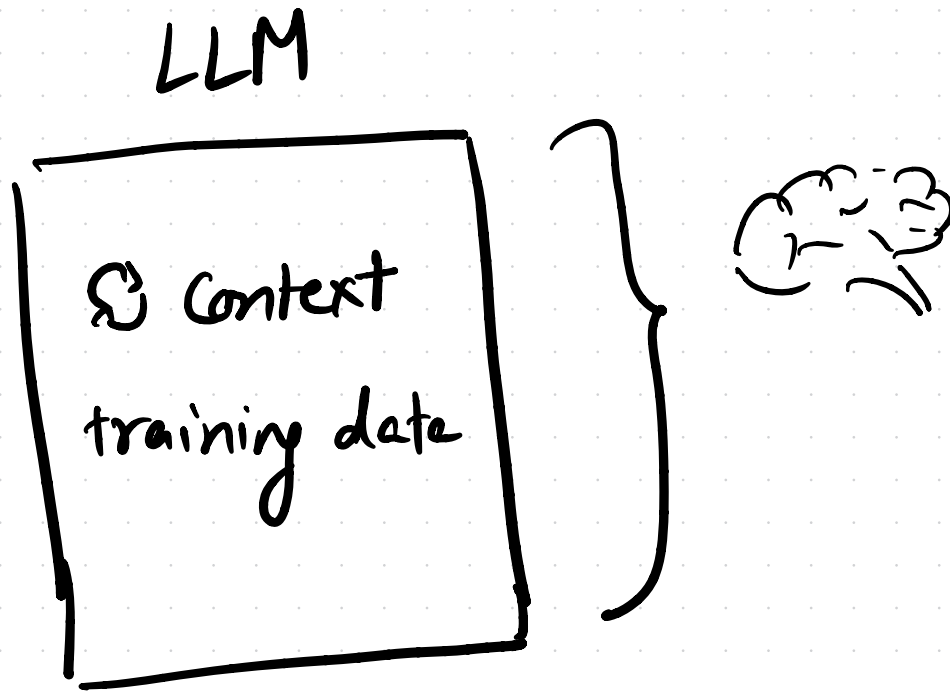
- ✓ Finding painting of sunset on a beach
- ✓ finding a similar dress
- ✓ Recommendations for similar hotels.

Popular Vector DBs

- Pinecone
- Weavate
- Chroma
- Redis Vector

RAG

(Realtime Augmented
Generation)



LLM's
Knowledge is
limited to its
training data

What is the latest
version of iPhone in
the market?



LLM



iPhone 15 Pro

(Hallucination)

When is my next
O'Reilly training?



LLM



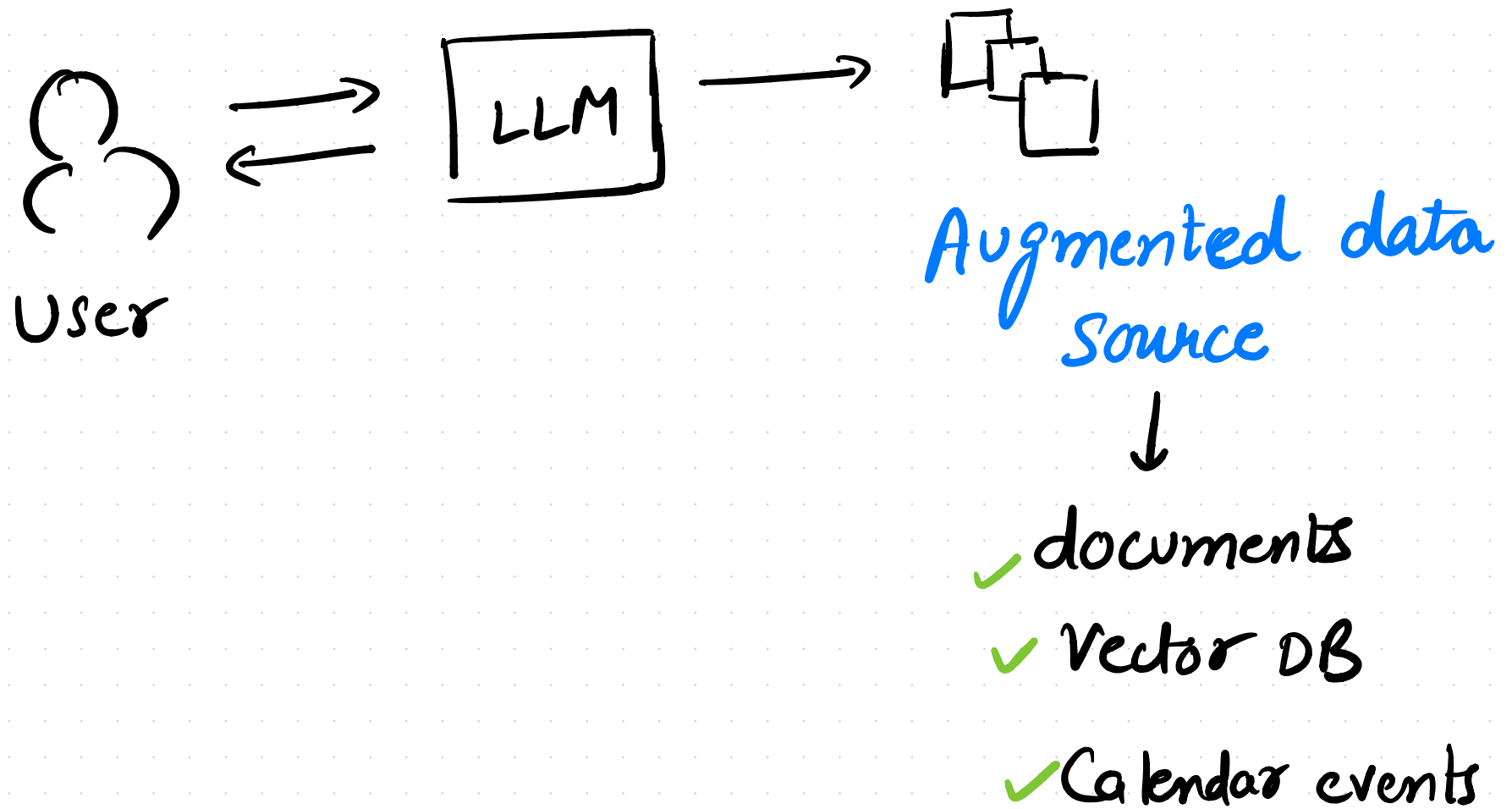
??

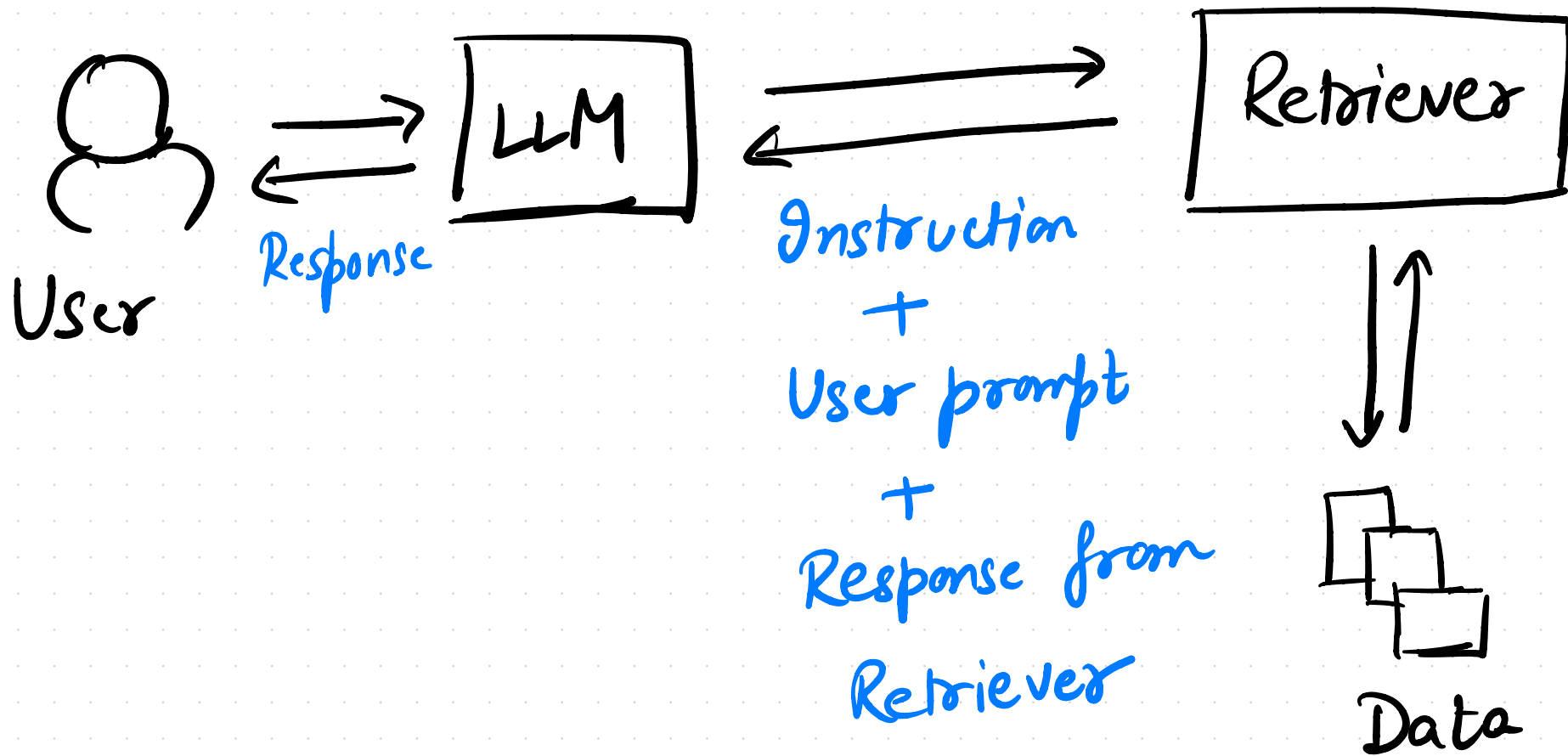
Challenges

① Out of date

② No source

How to extend
LLM's knowledge?





By extending the source
LLMs are less likely
to Hallucinate

Pros

- ✓ Latest information from external sources
- ✓ Extended knowledge without extending the size of LLM

Limitation

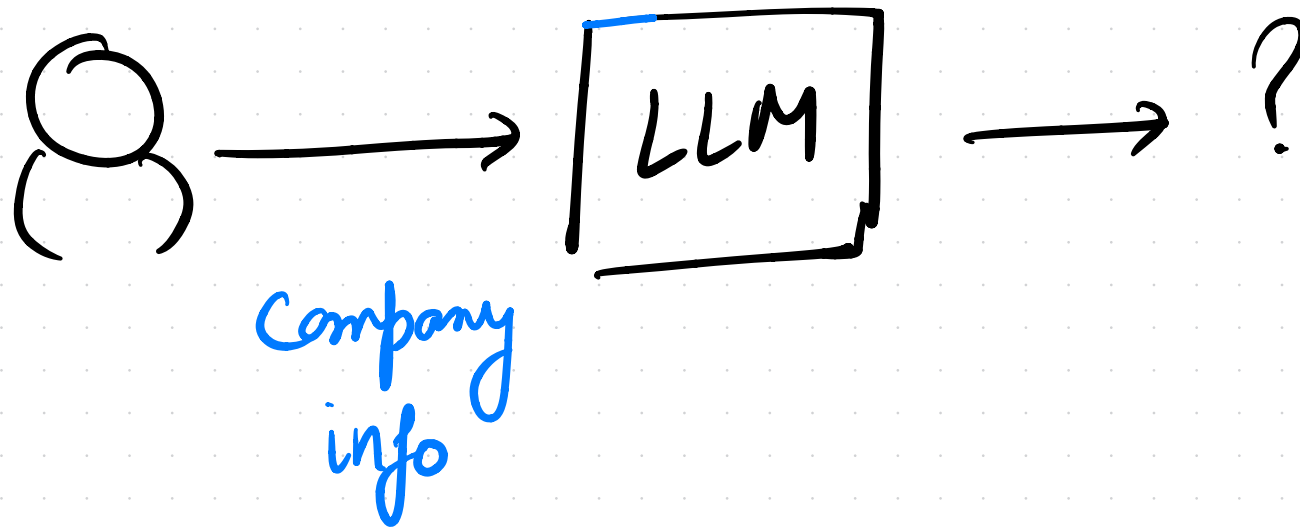
- Two step pipeline

Retriever + LLM $\rightarrow$ slower

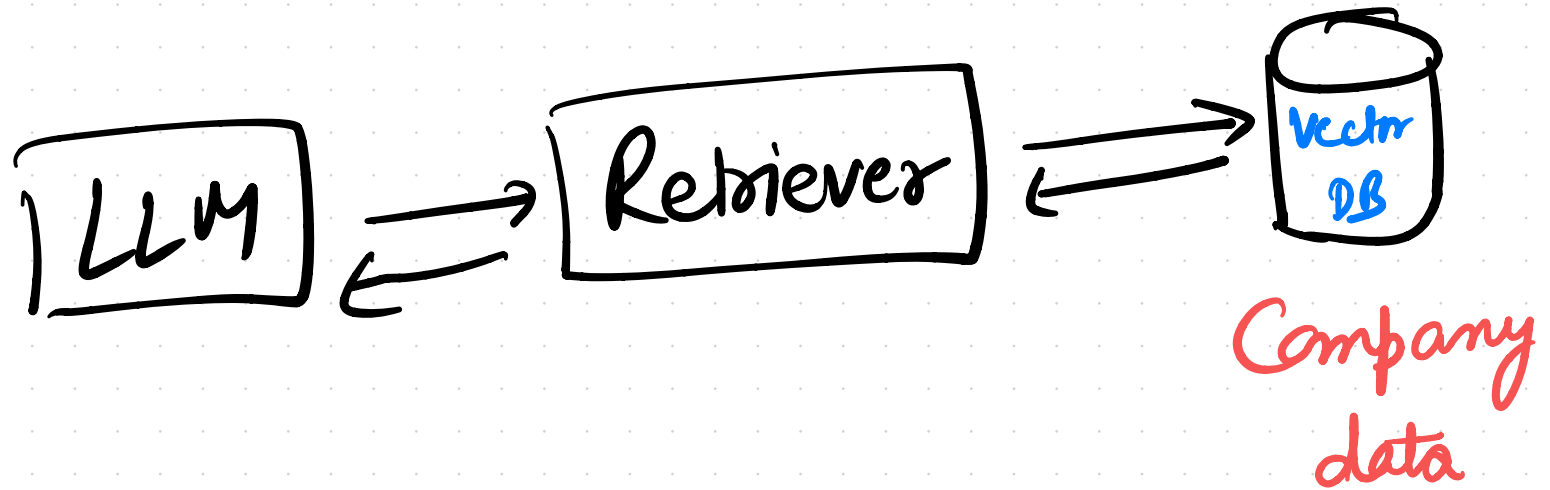
- Errors in Retrieval = bad answers

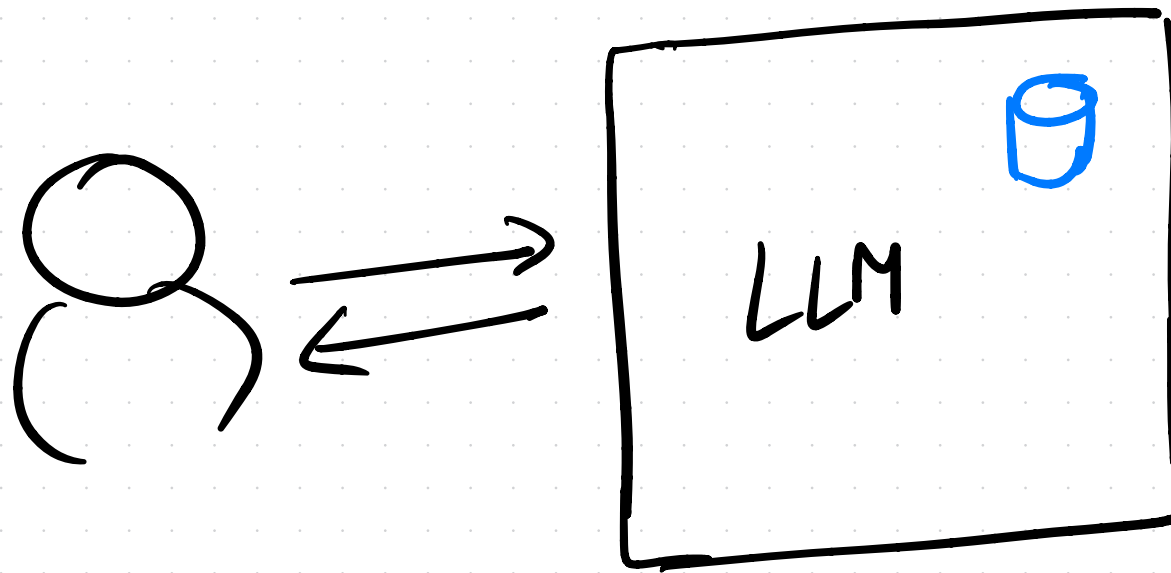
CAG

(Context Augmented
generation)



Using RAG





Provide Company
info directly
to LLM
Context

Instead of calculating data at
runtime, LLM is **trained / finetuned**
on a specific domain

Pros

✓ Faster

✓ Easy to implement on
Small knowledge sets

Limitation

- Context size is limited
- Needs frequent updating if data changes
- Difficult to scale on large data

AI Agents

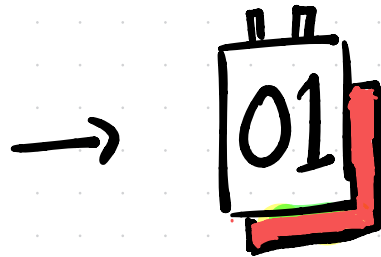
LLMs are limited

→ limited to the knowledge 

→ They are passive (wait for user action)

→ Can't Act

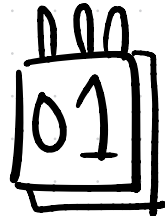
Create Event



AI Workflows

Fetch all my upcoming trainings
from Gmail



Add the events to my 



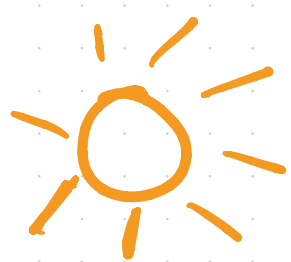
Generate table with columns

name		date/time		training link
------	--	-----------	--	---------------



Add to my Notes

Workflows extend capabilities of LLMs but they follow predefined paths set by humans



→ Weather?

AI workflow X

Agents

① Reasoning

- Break down the problem
- Make a plan

② Action

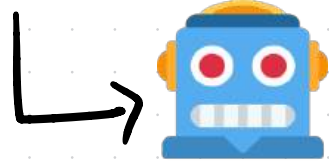
- Use external tools like Calendars, Calculators, DB

③ Iterate

- Iterate the steps if problem was not solved in one way

Plan → Act → Iterate

Schedule training
for Dec 5



→ check emails & calendar
for existing courses

→ check for course on
Dec 5
↓

Send conf
email to
produce

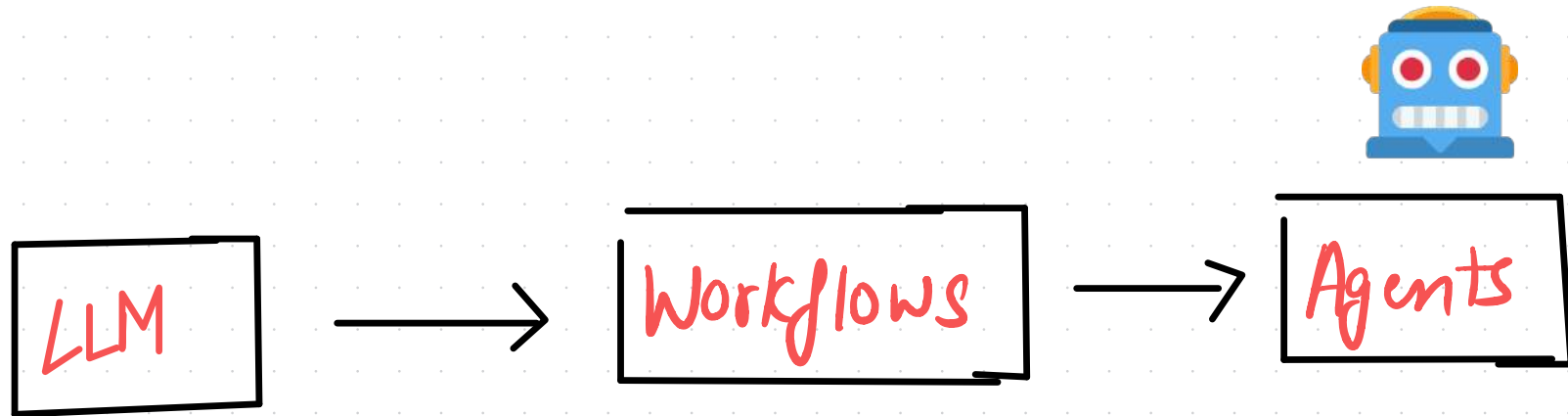
← Add event
to my
Calendar

← No course found

Agents are flexible, adaptive and well structured for open ended tasks

Workflows are efficient but follow a fixed path

Evolution

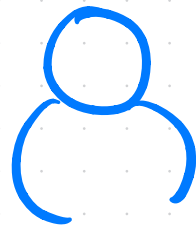


MCP

(Model Context Protocol)

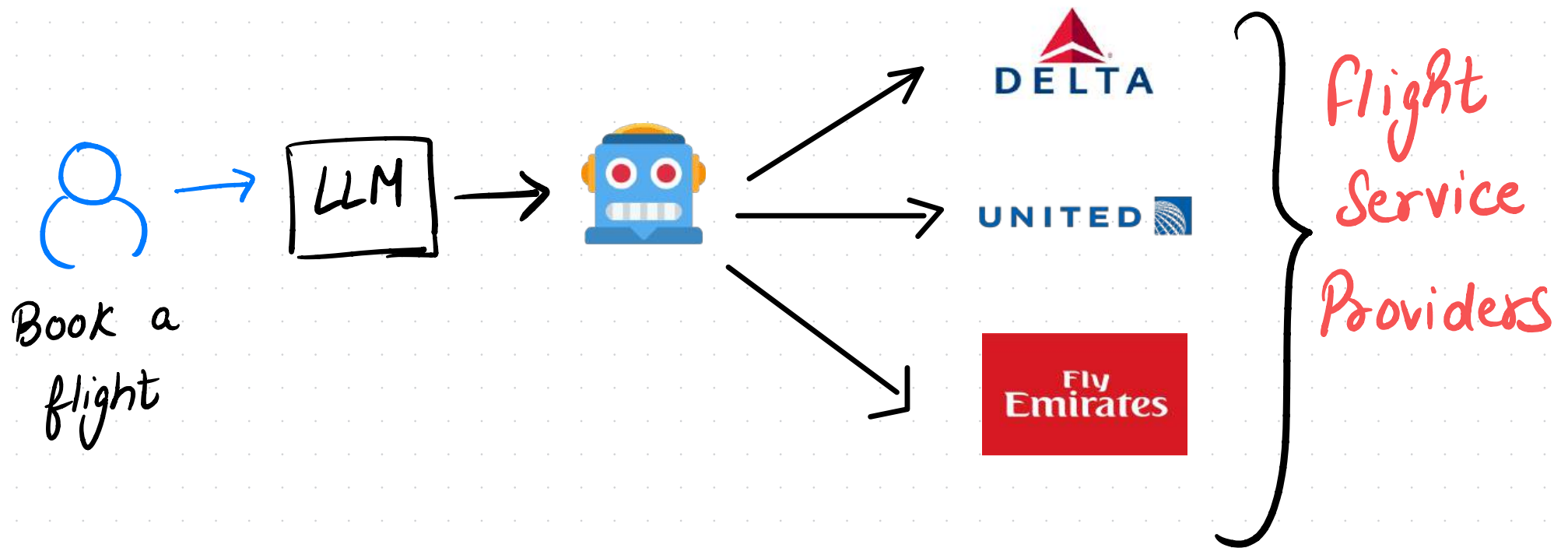


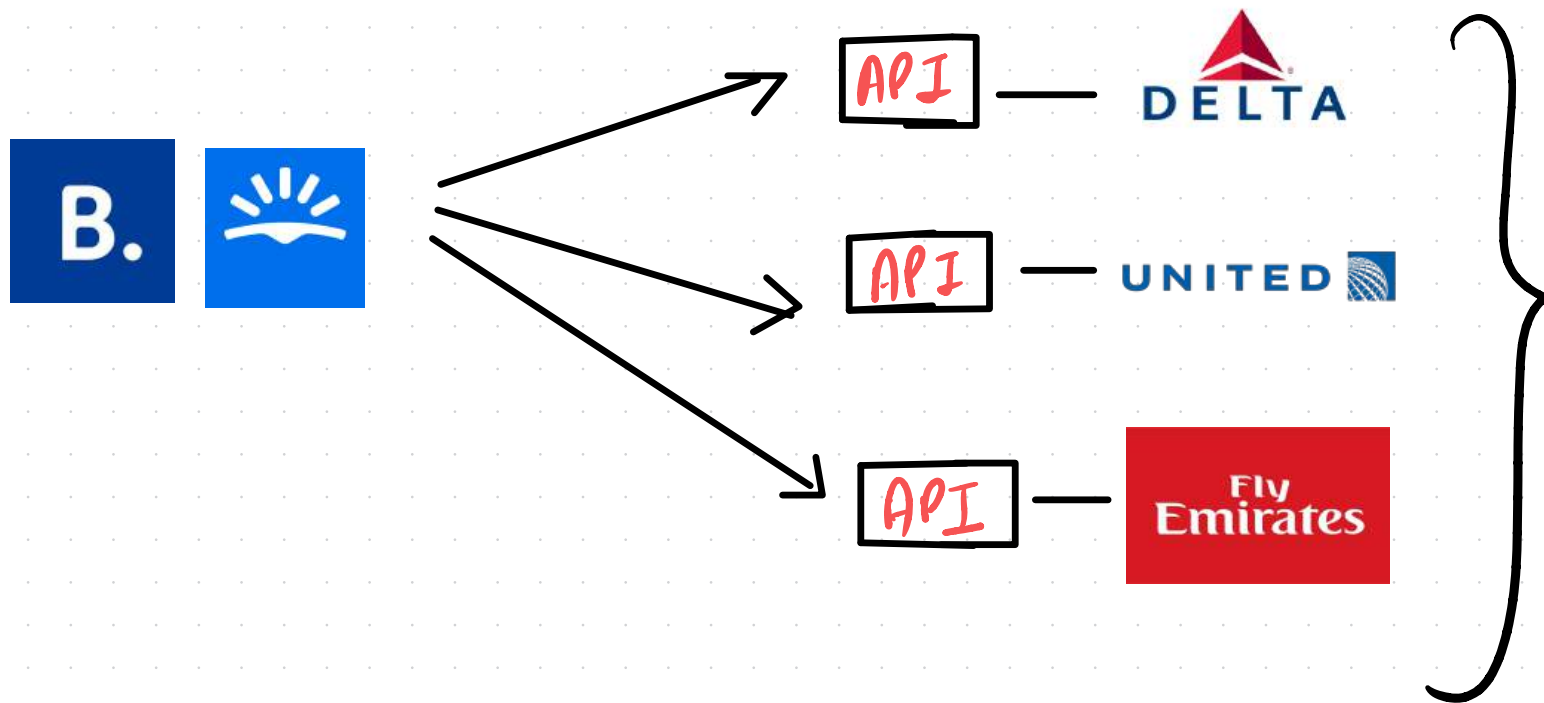
Booking



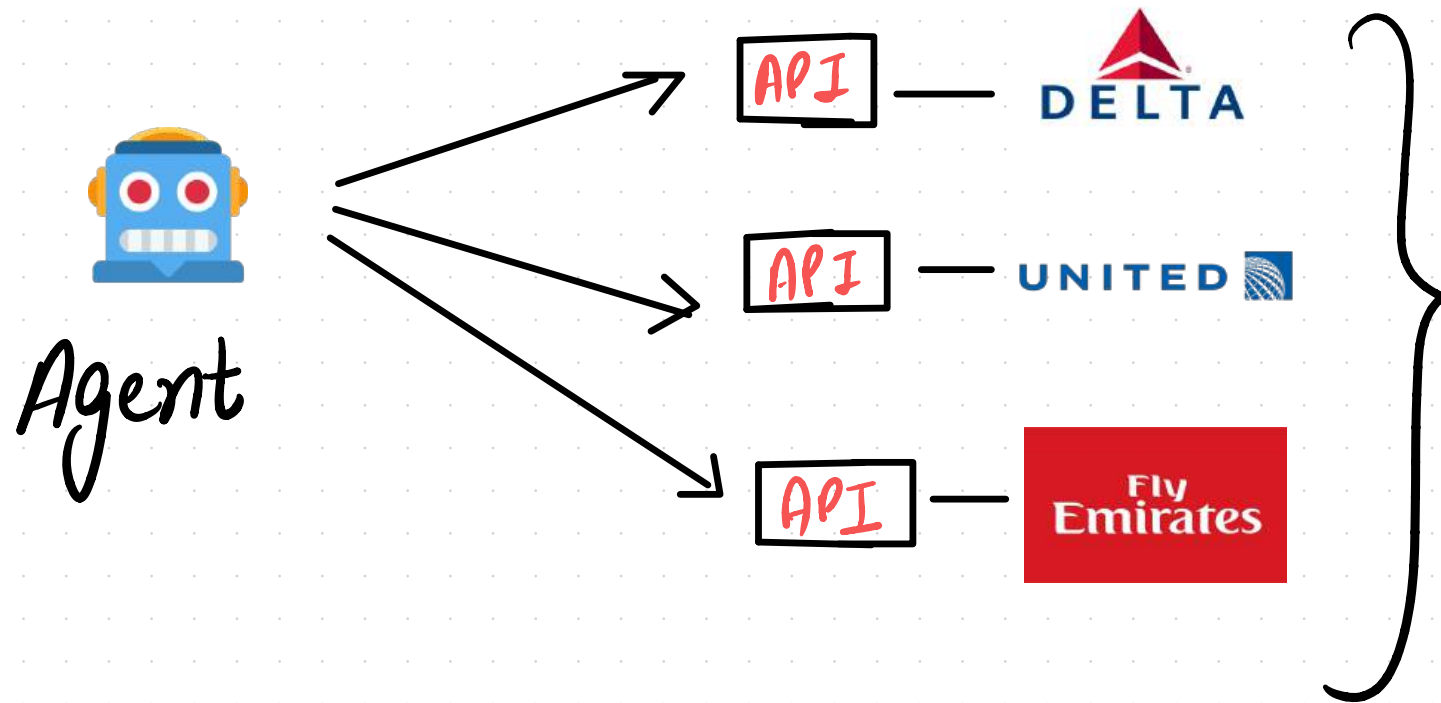
Book flight
from Chicago to
Amsterdam

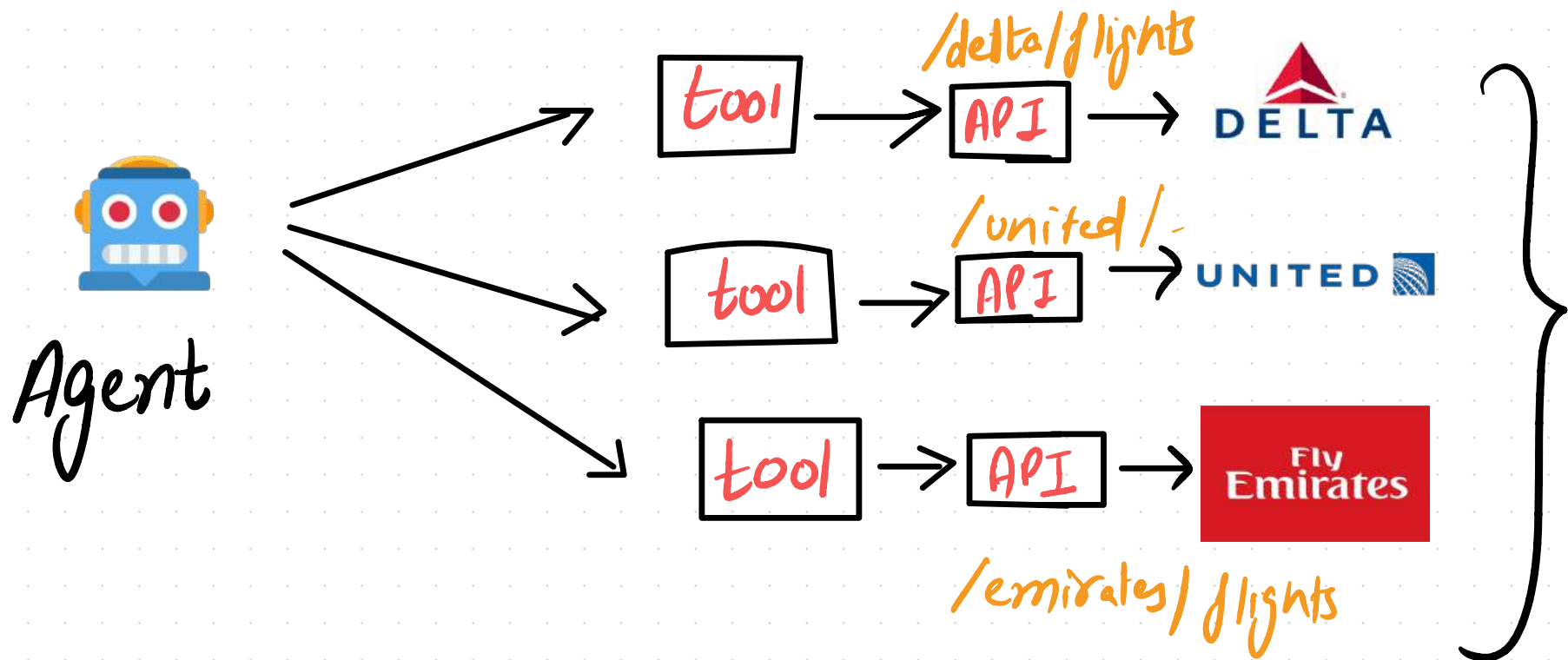
✗ LLM can't book
a flight but
only tells how to
book



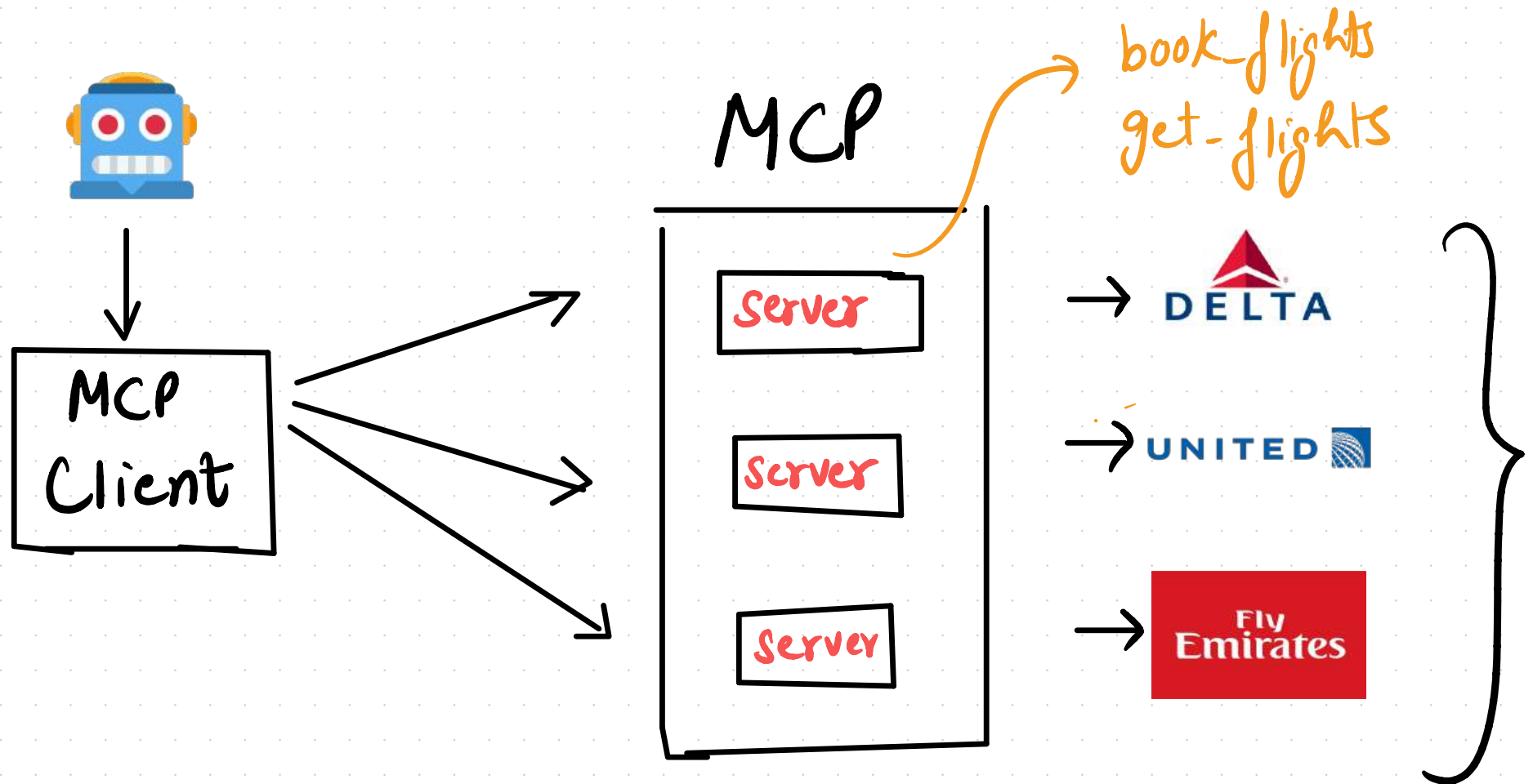


Instead of third party websites, we have Agents booking the flights

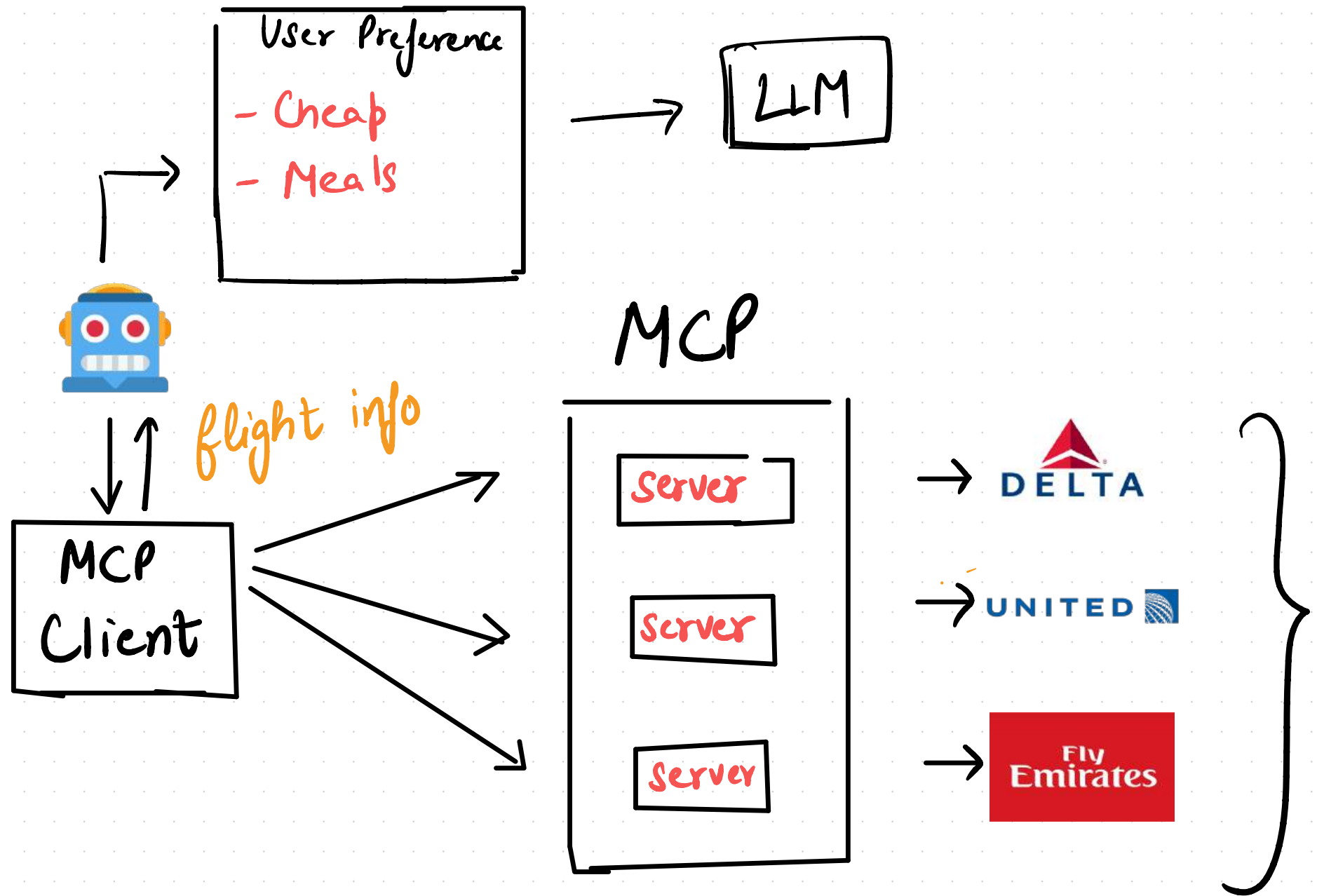




Agent will have to make
3 different API calls



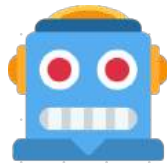
Each website will expose its own MCP server
MCP client will talk to MCP server to get
flight details



LLM



decides which
flight to book

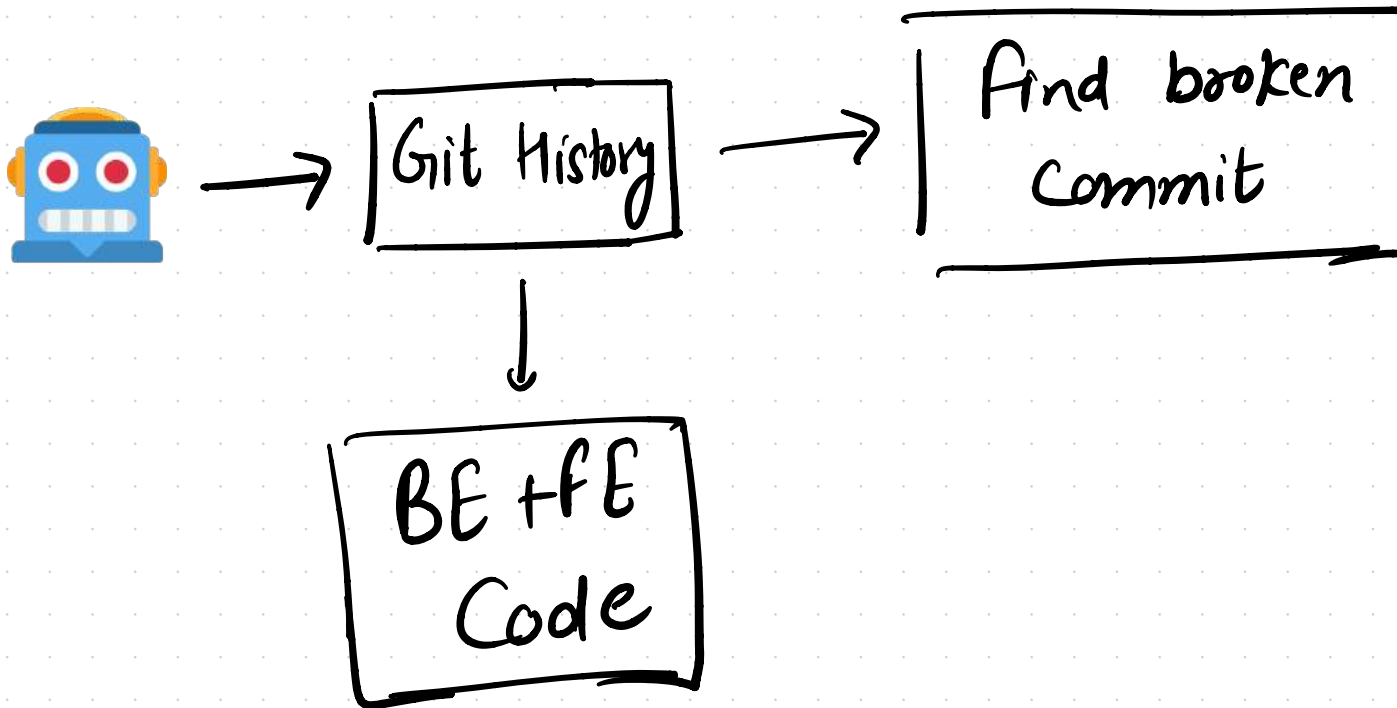


Book the flight

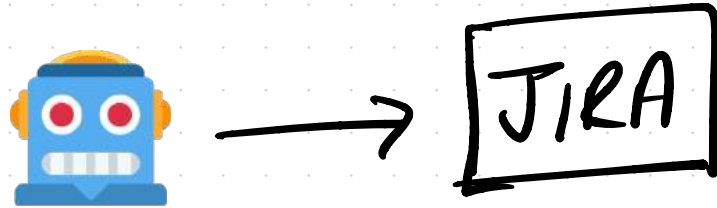
Agent

Real world Scenarios

→ Login functionality is broken on UI



JIRA MCP



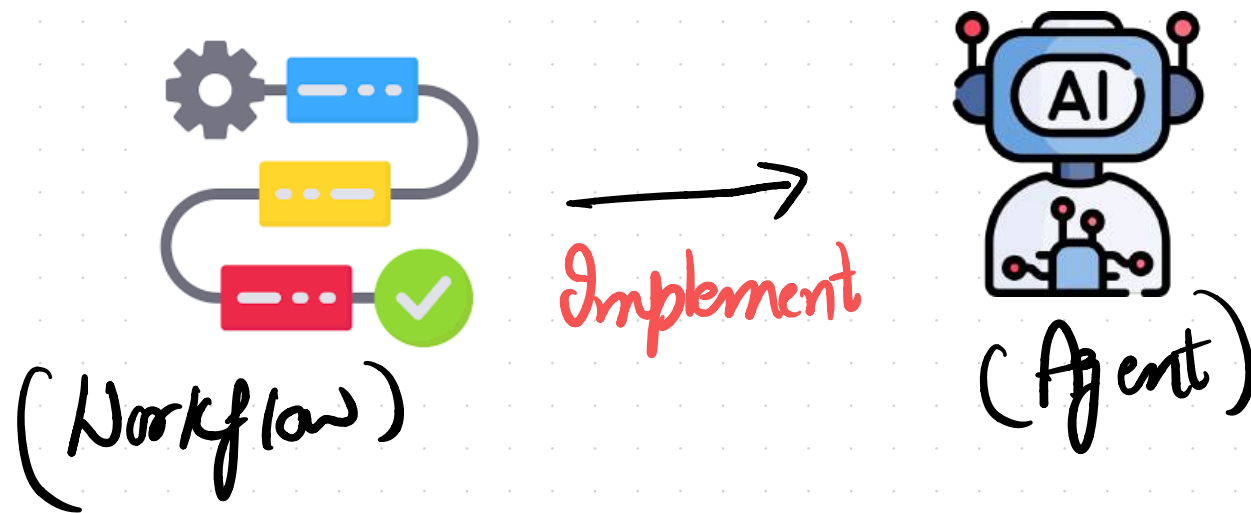
→ create issue

→ get issue

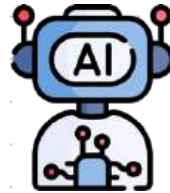
→ start to work on a ticket

SKILLS

Think about a workflow you repeatedly ask
your agent to implement

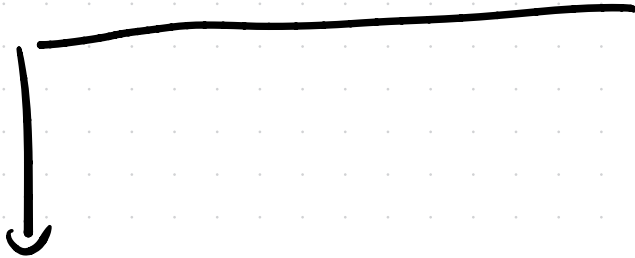


Create PR



Instruction

- Project id
- Reviewers
- PR description format



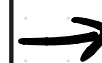
Create
branch



Commit



Push



PR description



Assign
reviewers

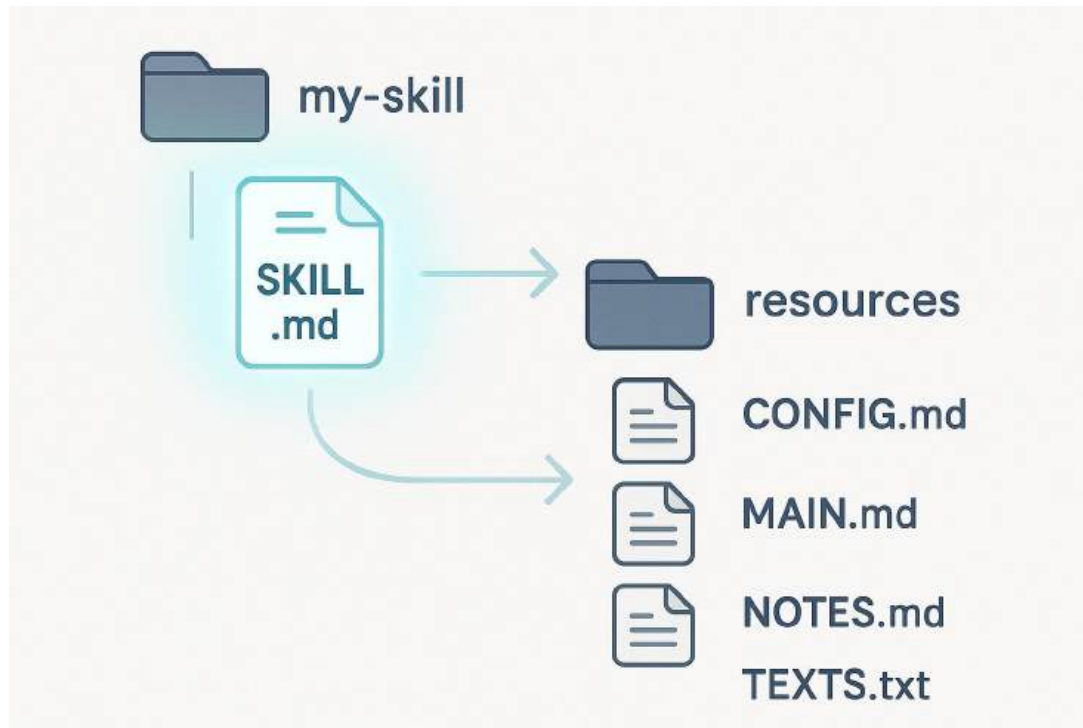
✗ Repeatedly writing long prompts everytime



Package everything into a skill



- Name
- Description
- Instructions



Agents can combine skills, MCPs
and subagents to create powerful
agentic workflows



Skills make workflows
deterministic as agents
exactly know what steps
to take

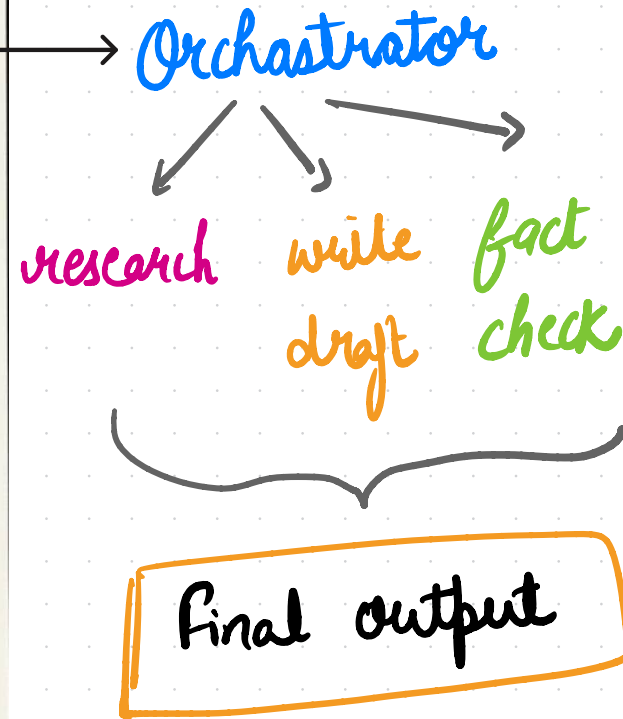
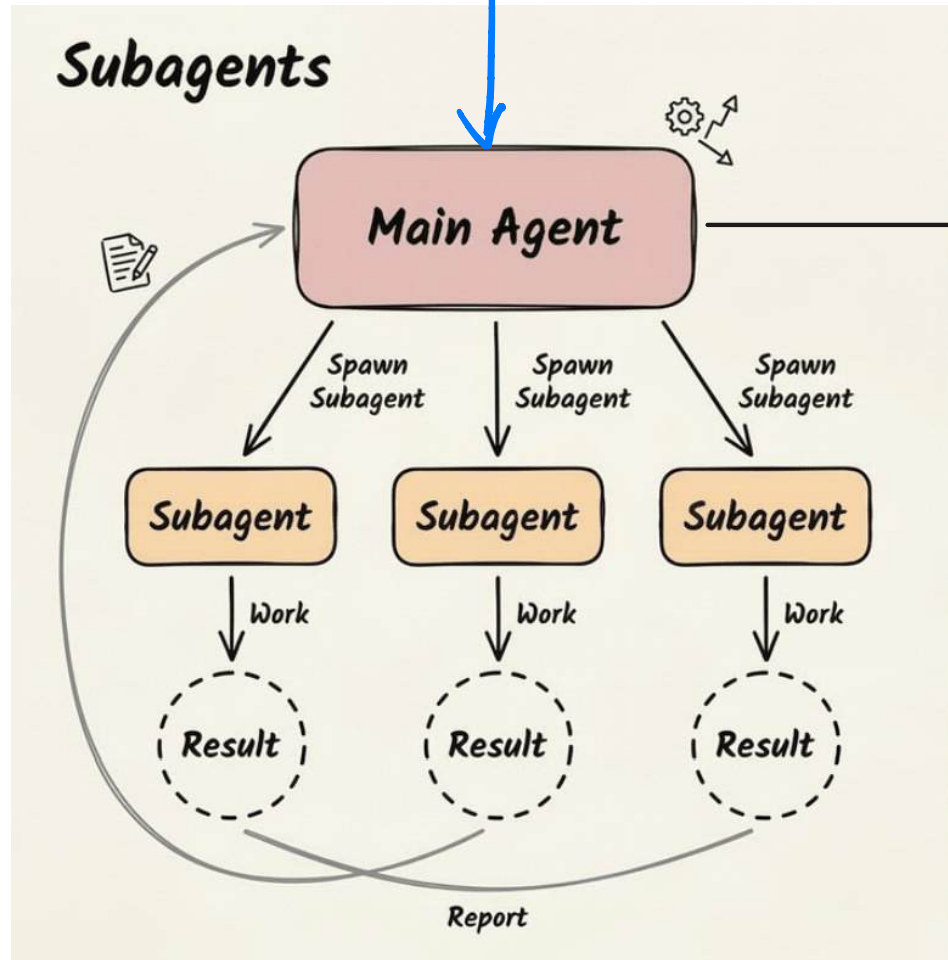
Subagents

Subagent is an agent spawned by another agent to handle a specific task



Every subagent has its own context window independent of the main Agent

User



Why Subagents ?

⇒ Parallelism

⇒ Context Management